



Agriya Yadav &lt;agriya.yadav\_ug2023@ashoka.edu.in&gt;

## Research Inquiry: CVaR RL for Arc Routing under Compound Distribution Shifts — Building on Your DRO Routing and SHIELD Work

15 messages

Agriya Yadav &lt;agriya.yadav\_ug2023@ashoka.edu.in&gt;

Mon, Apr 20, 2026 at 4:42 PM

To: "zgcao@smu.edu.sg" &lt;zgcao@smu.edu.sg&gt;

Dear Prof. Cao,

I am Agriya, an undergraduate in Computer Science and Mathematics at Ashoka University, India (graduating 2027). I am writing because my current project sits at the intersection of two threads your group has developed: distributionally robust optimization for routing (AAAI 2022) and multi-task multi-distribution VRP generalization (SHIELD, ICML 2025).

I am developing ShARC — a framework applying CVaR-based risk-sensitive RL to the Hierarchical Capacitated Arc Routing Problem (HARP) under simultaneous, continuous distribution shifts. The shift model is a compound vector  $\phi = (\delta_d, \delta_c, \delta_p)$  parameterising bidirectional drift across arc demands, traversal costs, and task availability probabilities. The key structural difference from the AAAI 2022 DRO routing formulation is that the uncertainty acts across three jointly-drifting parameters simultaneously rather than over a worst-case distribution on node coordinates — making the uncertainty set non-rectangular and non-decomposable.

The policy is a transformer-based encoder-decoder pointer network trained via CVaR-REINFORCE, with a shift curriculum that progressively increases  $\phi$  during training. I have a working implementation on OSM-derived benchmark instances with baselines (ILS, ACO, risk-neutral REINFORCE) running on eval data now.

Two specific questions I would value your perspective on: (1) the SHIELD framework handles multi-distribution generalization via sparsity and hierarchical decoding — I am curious whether a similar structural inductive bias would help under continuously-drifting rather than discretely-enumerated distributions; and (2) how you see CVaR-regret framing as a theoretical contribution distinct from the distributional robustness framing in your DRO routing paper.

I have prior publication experience (AAAI ICWSM) and the project is being developed with a faculty collaborator at Ashoka, targeting AAAI 2027. Your website mentions remote interns are welcome — I would be interested in that possibility as well if there is a fit.

Best regards,

Agriya

B.Sc. CS &amp; Mathematics, Ashoka University

[website](#) | [linkedin](#)

CAO Zhiguang &lt;zgcao@smu.edu.sg&gt;

Mon, Apr 20, 2026 at 7:12 PM

To: Agriya Yadav &lt;agriya.yadav\_ug2023@ashoka.edu.in&gt;

SMU Classification: Restricted

Hi Agriya,

Thanks for reaching out. Regarding 'I have a working implementation', you already got some results? Regarding your questions, I agree that structural inductive bias could be useful under continuously drifting distributions, although it may need to be designed differently from the discretely enumerated setting in SHIELD. In particular, sparsity or hierarchical decoding may help if the shift parameters induce local or structured changes in routing decisions. For the CVaR-regret framing, I think the key distinction is that it emphasizes tail-risk-sensitive performance degradation under shifts, while classical DRO is usually framed around worst-case expected performance over an uncertainty set. This distinction could be valuable if you can clearly show both theoretical and empirical advantages. However, I may advise further after seeing the empirical results.

Best regards,

Zhiguang

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**Agriya Yadav** <agriya.yadav\_ug2023@ashoka.edu.in>  
To: CAO Zhiguang <zgcao@smu.edu.sg>

Mon, Apr 20, 2026 at 9:49 PM

Dear Prof. Cao,

Thank you — this is very helpful. Your point about the CVaR-regret vs DRO distinction needing both theoretical and empirical support is exactly the bar I am trying to clear.

The implementation is complete and baselines are currently running on the held-out eval set. I expect preliminary results within the next few days. I will follow up with the cost-vs-shift-severity curves once they are ready, and I would be very grateful for your read on whether the empirical signal is strong enough to support the theoretical framing.

Best regards,  
Agriya

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**Agriya Yadav** <agriya.yadav\_ug2023@ashoka.edu.in>  
To: CAO Zhiguang <zgcao@smu.edu.sg>

Thu, Apr 23, 2026 at 2:44 PM

Dear Prof. Cao,

Following up as promised — please find attached the preliminary cost-vs-shift-severity curves from the held-out eval set (N=60 graphs, V=2 vehicles, 5 seeds per severity level).

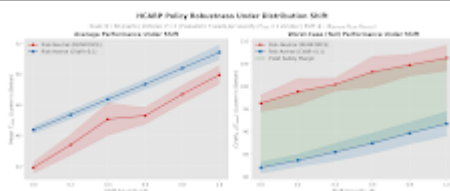
The key result is in the right panel: the CVaR-trained policy achieves a consistently lower worst-case tail cost ( $\text{CVaR}_{0.1}$  of  $T_{\max}$ ) across all shift severities  $\phi \in [0, 1]$ , with the safety margin widening as the shift grows more severe. The left panel shows the expected mean-tail tradeoff — the risk-averse policy is more conservative on average at low-to-mid severities, which I take as a sanity check that the CVaR-REINFORCE gradient estimator is behaving correctly rather than as a weakness.

I think this provides the empirical grounding for the CVaR-regret vs DRO distinction you identified: the tail protection is a genuine gain under compound shift, not reducible to optimising worst-case expected loss. That said, I would value your read on whether the signal is strong enough to support that framing, or whether you see gaps before this is publication-ready.

ILS and ACO baselines are still running and I will share those results once complete.

Best regards,  
Agriya

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severity\_plot\_v3.png  
481K

**Agriya Yadav** <agriya.yadav\_ug2023@ashoka.edu.in>  
To: CAO Zhiguang <zgcao@smu.edu.sg>

Mon, Apr 27, 2026 at 5:44 AM

Hi Professor,

Just following up.

Sincerely,

Agriya

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**CAO Zhiguang** <zgcao@smu.edu.sg>  
To: Agriya Yadav <agriya.yadav\_ug2023@ashoka.edu.in>

Mon, Apr 27, 2026 at 9:27 AM

SMU Classification: Restricted

Hi Agriya

The right panel looks good. From what I observed, AAAI may like more general problems like considering multiple VRP variants at the same time, to show the approach is generic. HCARP might be too specific, or if you still have different variants of it. If you only have this single variant, you may argue those VRP variants in AI conference like TSP, CVRP, PDP, etc, are not realistic or challenging enough. Also, why you focus on robust attribute. Moreover, please do as many experiments as possible. I have never tried ICDE, but I have a feeling ICDE pay prefer a specific problem using real data (like your OSM) over AAAI. But again, that is only my intuition, you and your supervisor know this work better. Btw, who is your supervisor?

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**Agriya Yadav** <agriya.yadav\_ug2023@ashoka.edu.in>  
To: CAO Zhiguang <zgcao@smu.edu.sg>

Mon, Apr 27, 2026 at 1:19 PM

Dear Prof. Cao,

Thank you for the feedback — this is exactly the kind of direction I needed.

My supervisor on this project is Prof. Bhargab B. Bhattacharya (ISI Kolkata / Ashoka), whose expertise is primarily on the combinatorial and graph-theoretic side of arc routing. The RL and distributional robustness framing is my own contribution, which is partly why I reached out — I have been looking for a collaborator with deeper expertise in that direction to sharpen the theoretical contributions and broaden the experimental scope.

On the generality concern: you are right that a single HCARP variant is a limitation for AAAI. I think the natural extension is to show that the CVaR-REINFORCE training strategy with compound shift curriculum transfers to standard VRP variants (CVRP, PDP) with minimal modification — the shift parameterisation  $\phi$  generalises straightforwardly to node-based problems. I am willing to run those experiments if that is what it takes to make the generality argument credible.

On venue: I had AAAI 2027 in mind, but I take your point about ICDE being a potentially better fit for the OSM real-data angle. I would value your intuition on which framing — generality across VRP variants for AAAI, vs. real-data specificity for ICDE — is the stronger paper given the current results.

Best regards,  
Agriya

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**CAO Zhiguang** <zgcao@smu.edu.sg>  
To: Agriya Yadav <agriya.yadav\_ug2023@ashoka.edu.in>

Mon, Apr 27, 2026 at 1:32 PM

SMU Classification: Restricted

One more thing, which is also very important, what is the novelty of your approach? Only because it has never been applied to VRP? In general, direct application of an approach to a new task is not enough for a top AI conference. Please spend more time on this aspect as well.

Best regards,  
Zhiguang

**From:** Agriya Yadav <agriya.yadav\_ug2023@ashoka.edu.in>

**Sent:** Monday, 27 April 2026 3:50 pm

**To:** CAO Zhiguang <zgcao@smu.edu.sg>

**Subject:** Re: Research Inquiry: CVaR RL for Arc Routing under Compound Distribution Shifts — Building on Your DRO Routing and SHIELD Work

Dear Prof. Cao,

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**Agriya Yadav** <agriya.yadav\_ug2023@ashoka.edu.in>

Mon, Apr 27, 2026 at 1:40 PM

To: CAO Zhiguang <zgcao@smu.edu.sg>

Dear Prof. Cao,

This is exactly the right question to push on, and I want to answer it precisely.

The novelty is not "CVaR applied to VRP" — I agree that would not be enough. The claim is a specific combination that no prior work has addressed:

1. Compound shift model: the uncertainty set is over  $\phi = (\delta_d, \delta_c, \delta_p)$  jointly — demand, cost, and availability drifting simultaneously. This makes the uncertainty set non-rectangular and non-decomposable, which is structurally different from DRO-VRP (uncertainty over node coordinate distributions) and SHIELD (discrete enumerated families). The compound structure requires a different gradient estimator and training curriculum.
2. Shift-context-conditioned decoder: the policy observes the current shift vector at inference time and adapts its routing decisions to the type and severity of the shift — not just averaging over shifts during training. No existing NCO architecture does this.
3. Within-episode budget signal: the running worst-case vehicle time  $\beta_t$  is included in the decoder context, giving the policy an explicit risk budget signal during rollout. This enables risk-sensitive arc selection within an episode, not just at training time.

The intersection of arc routing + continuous compound shift + shift-conditioned policy + CVaR tail objective is unoccupied in the literature. I have a detailed architecture comparison document with a literature table I can share if useful. On your earlier point about generality — I am also planning to show that the shift curriculum and CVaR-REINFORCE modifications transfer to CVRP/PDP with minimal changes(hopefully), which would address the AAAI reviewers' likely concern about problem-specificity.

Best regards,

Agriya

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**Agriya Yadav** <agriya.yadav\_ug2023@ashoka.edu.in>

Fri, May 1, 2026 at 2:04 PM

To: CAO Zhiguang <zgcao@smu.edu.sg>

Following up on whether this makes sense professor.

Sincerely,

Agriya

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**CAO Zhiguang** <zgcao@smu.edu.sg>

Fri, May 1, 2026 at 10:43 PM

To: Agriya Yadav <agriya.yadav\_ug2023@ashoka.edu.in>

SMU Classification: Restricted

2. 'the policy observes the current shift vector at inference time' - how to measure those shift vector, e.g., the shift of cost?

3. 'the running worst-case vehicle time  $\beta_t$  is included in the decoder context' - worst-case vehicle time  $\beta_t$  of the same instance you are inferring? If yes, how can you get  $\beta_t$  before you start to solve this instance?

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**Agriya Yadav** <agriya.yadav\_ug2023@ashoka.edu.in>  
To: CAO Zhiguang <zgcao@smu.edu.sg>

Wed, May 6, 2026 at 2:52 PM

Dear Prof. Cao,

Apologies for the delayed reply — I was in the middle of my final examination period and wanted to give these questions the careful answer they deserve rather than rush a response.

Both questions point at a real implementation detail, so let me be precise.

On question 2 (measuring the shift vector at inference time): the shift vector  $\phi = (\delta_d, \delta_c, \delta_p)$  is not inferred from the instance — it is observed as a context input, similar to how SHIELD conditions on a task embedding. In deployment, the assumption is that the operating environment can characterise its current distributional regime (e.g., from historical sensor logs, demand forecasts, or a monitoring system), and  $\phi$  is provided to the policy as a descriptor of that regime. At evaluation time in our experiments,  $\phi$  is known because we explicitly sample it to construct the shifted instance — we are evaluating under a controlled shift rather than trying to estimate it from data. I agree this is a meaningful practical assumption and worth stating clearly as a limitation or future direction (online shift estimation). The contribution is the policy architecture and training procedure given that  $\phi$  is observed, not a method for estimating  $\phi$ .

On question 3 ( $\beta_t$  before solving the instance):  $\beta_t$  is not the ground-truth worst-case time — it is the running maximum vehicle time accumulated so far during the current rollout, i.e.,  $\max_{\{k \leq t\}} T_k$  where  $T_k$  is the completion time of vehicle  $k$  after its  $k$ -th arc assignment. It is updated incrementally as the policy assigns arcs step-by-step, so at each decoding step  $t$  the policy has access to the worst-case vehicle time achieved among decisions already made. This is analogous to a running budget signal: the policy can "see" how imbalanced the current partial solution is and adjust subsequent arc assignments accordingly. There is no circularity —  $\beta_t$  at step  $t$  is fully determined by the partial assignments made in steps 1 through  $t$ .

I can share the decoder architecture diagram and pseudocode if that would make these mechanisms clearer.

Best regards,  
Agriya

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**Agriya Yadav** <agriya.yadav\_ug2023@ashoka.edu.in>  
To: CAO Zhiguang <zgcao@smu.edu.sg>

Wed, May 13, 2026 at 8:58 AM

Hi Professor,

Just following up.

Sincerely,  
Agriya

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**CAO Zhiguang** <zgcao@smu.edu.sg>  
To: Agriya Yadav <agriya.yadav\_ug2023@ashoka.edu.in>

Wed, May 13, 2026 at 9:04 AM

SMU Classification: Restricted

I would like to comment further after I have seen the draft with numeric results. Thanks for your understanding.

Best regards,  
Zhiguang

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**From:** Agriya Yadav <agriya.yadav\_ug2023@ashoka.edu.in>

**Sent:** Wednesday, 13 May 2026 11:28 am

**To:** CAO Zhiguang <zgcao@smu.edu.sg>

**Subject:** Re: Research Inquiry: CVaR RL for Arc Routing under Compound Distribution Shifts — Building on Your DRO Routing and SHIELD Work

Hi Professor,

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**Agriya Yadav** <agriya.yadav\_ug2023@ashoka.edu.in>

To: Agriya Yadav <Agriyayadav@gmail.com>

Wed, May 13, 2026 at 5:41 PM

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